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Basics of Windows Server and Active Directory Theory :Overview of Windows Server and its roles. Introduction to Active Directory (AD): Concepts, architecture, and uses. Practical :Install Windows Server and configure Active Directory Domain Services (AD DS).Task: Submit a document on the steps to install and configure AD DS.

1. Install Windows Server on a virtual machine using VirtualBox or VMware and take a screenshot of the desktop after the installation completes.

ANS:

Step 1: Create a New Virtual Machine

1. Open VMware Workstation or VirtualBox.
2. Click on Create a New Virtual Machine.
3. Select the Windows Server 2019 ISO file.
4. Select the operating system as Microsoft Windows.
5. Choose Windows Server 2019 (64-bit).
6. Assign RAM and CPU resources.
7. Create a virtual hard disk of at least 60 GB.
8. Finish the virtual machine configuration.

Step 2: Start Windows Server Installation

1. Start the virtual machine.
2. Select the language, time, and keyboard settings.
3. Click Install Now.
4. Select Windows Server 2019 Standard Evaluation (Desktop Experience).
5. Accept the license agreement.
6. Select Custom: Install Windows Only.
7. Select the virtual hard disk.
8. Click Next and wait for the installation to complete.

Step 3: Set Administrator Password

1. After installation, the server will restart.
2. Set a strong password for the Administrator account.
3. Press Ctrl + Alt + Delete (send the command through the VM menu if necessary).
4. Log in using the Administrator account.

Result

Windows Server 2019 was successfully installed on the virtual machine, and the desktop is displayed.

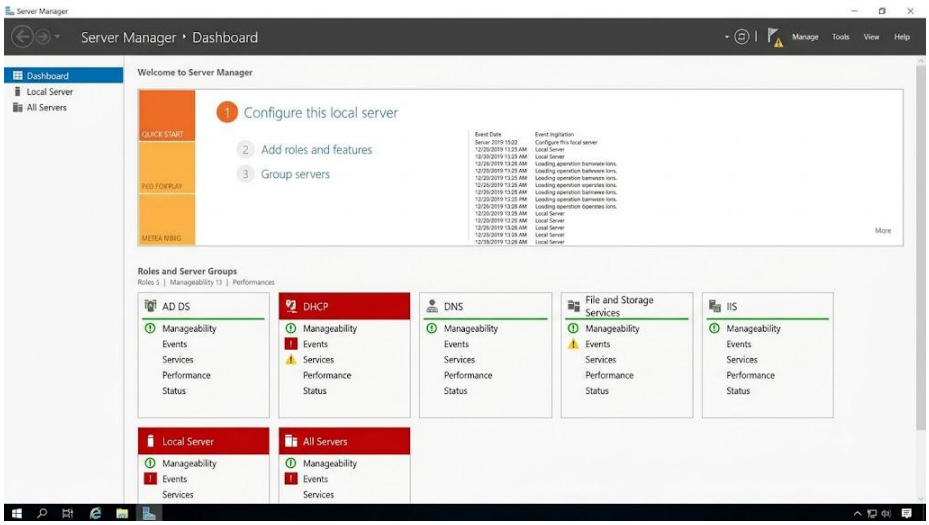


2. Open Server Manager and use the 'Add roles and features' wizard to install the Active Directory Domain Services (AD DS) role. Document each step with a short description and screenshot.

ANS:

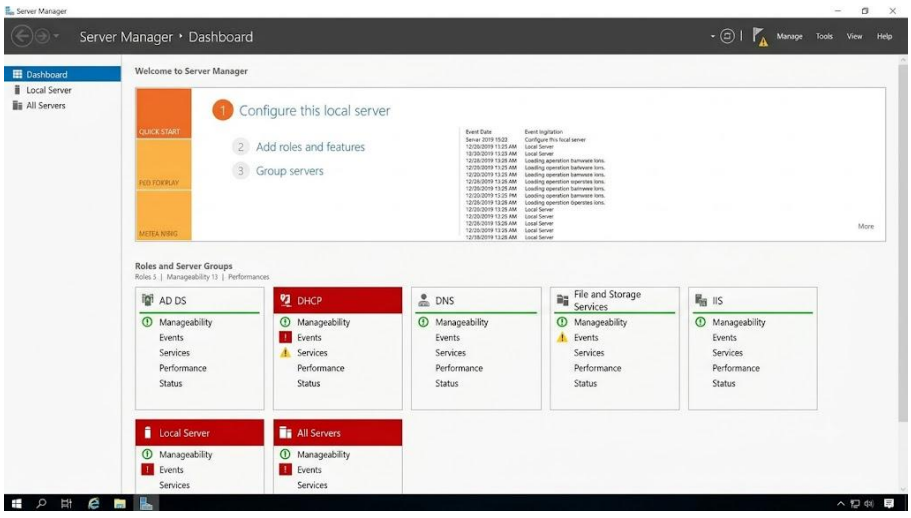
Step 1: Open Server Manager

1. Log in to Windows Server using the Administrator account.
2. Open Server Manager.
3. Click Manage from the top-right corner.
4. Select Add Roles and Features.



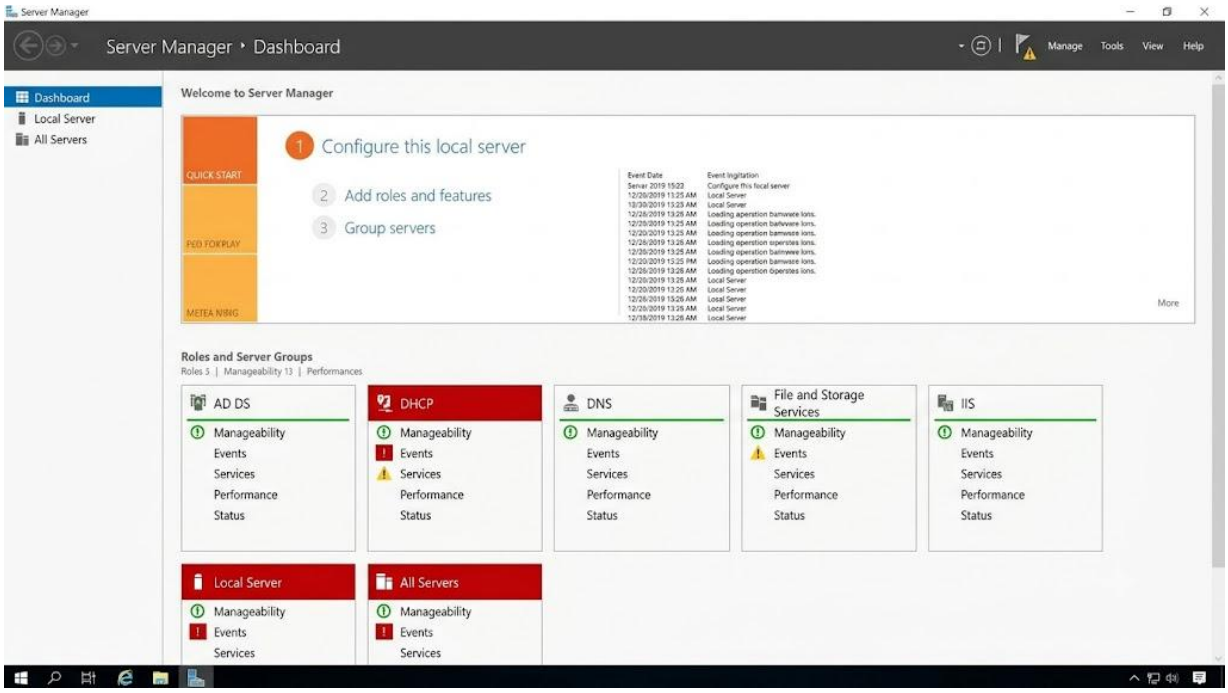
Step 2: Before You Begin

1. The Add Roles and Features Wizard will open.
2. Read the introduction.
3. Click Next.



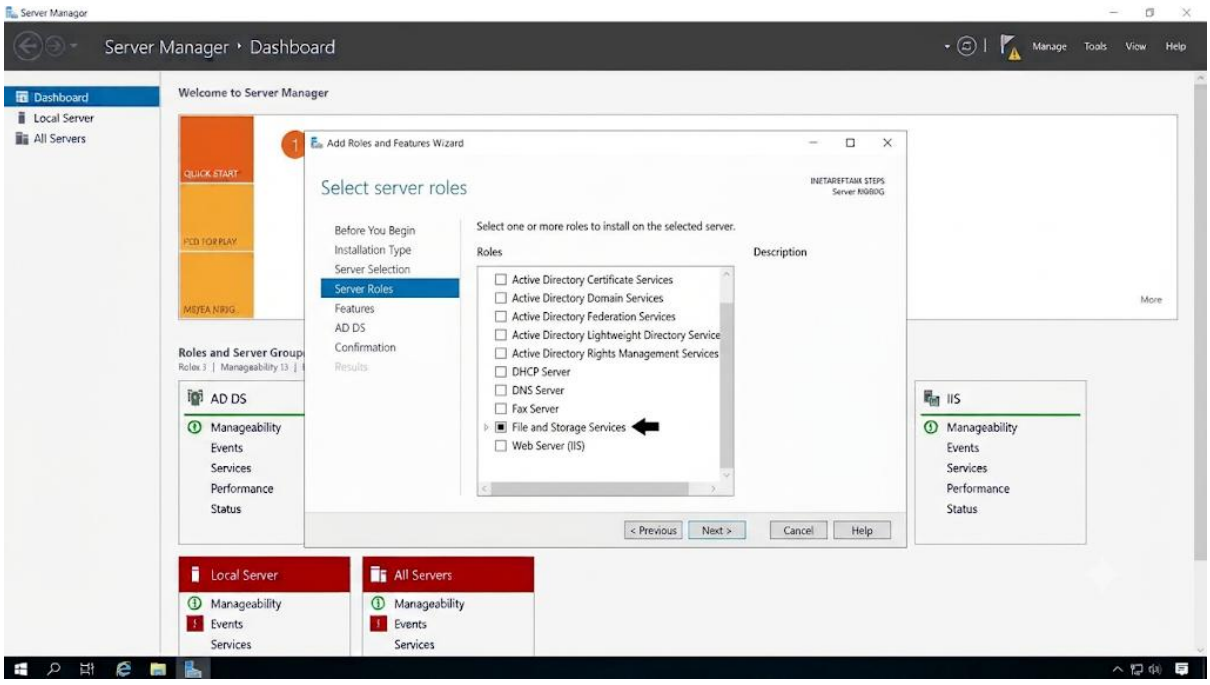
Step 3: Select Installation Type

1. Select Role-based or feature-based installation.
2. Click Next.



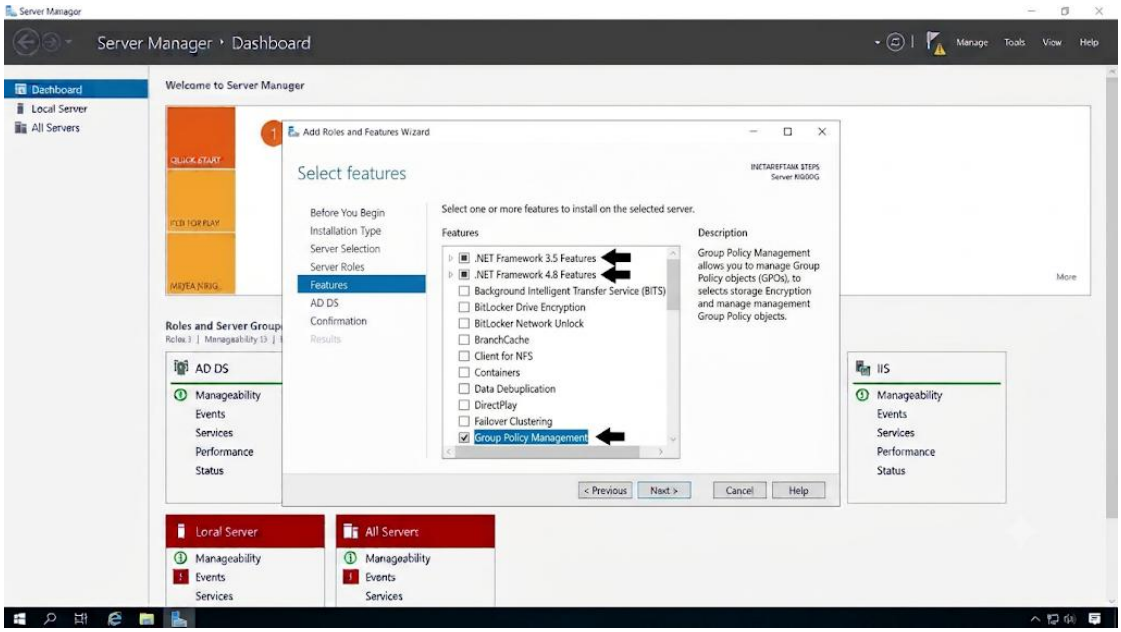
Step 4: Select Destination Server

1. Select Select a server from the server pool.
2. Select your local Windows Server.
3. Click Next.



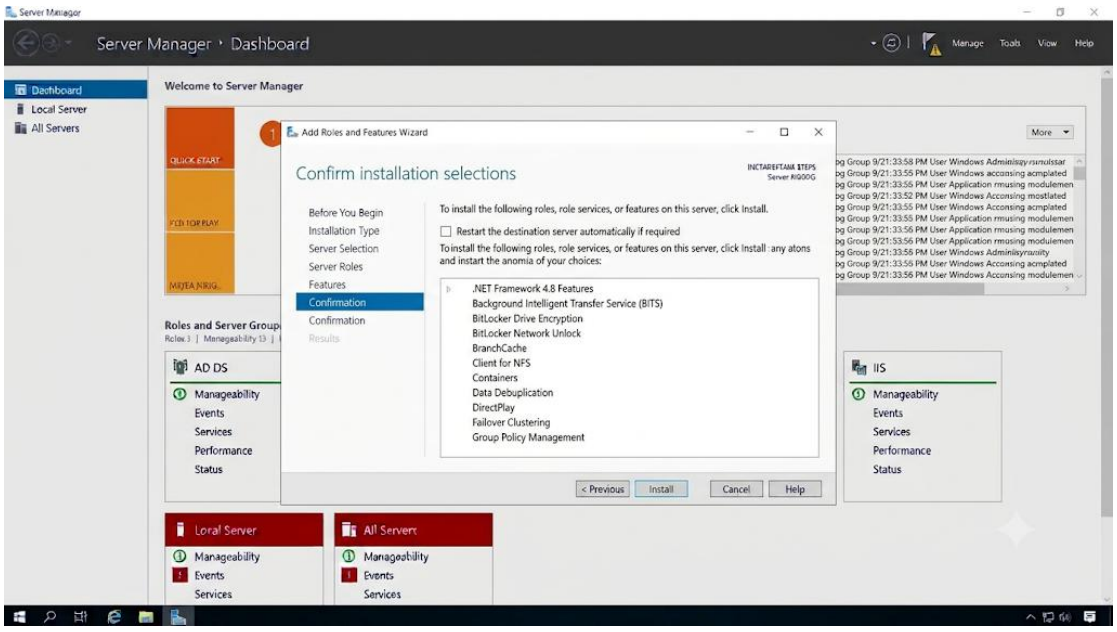
Step 5: Select Server Roles

1. Find Active Directory Domain Services.
2. Check the box beside Active Directory Domain Services.
3. A pop-up window will appear.
4. Click Add Features.
5. Click Next.



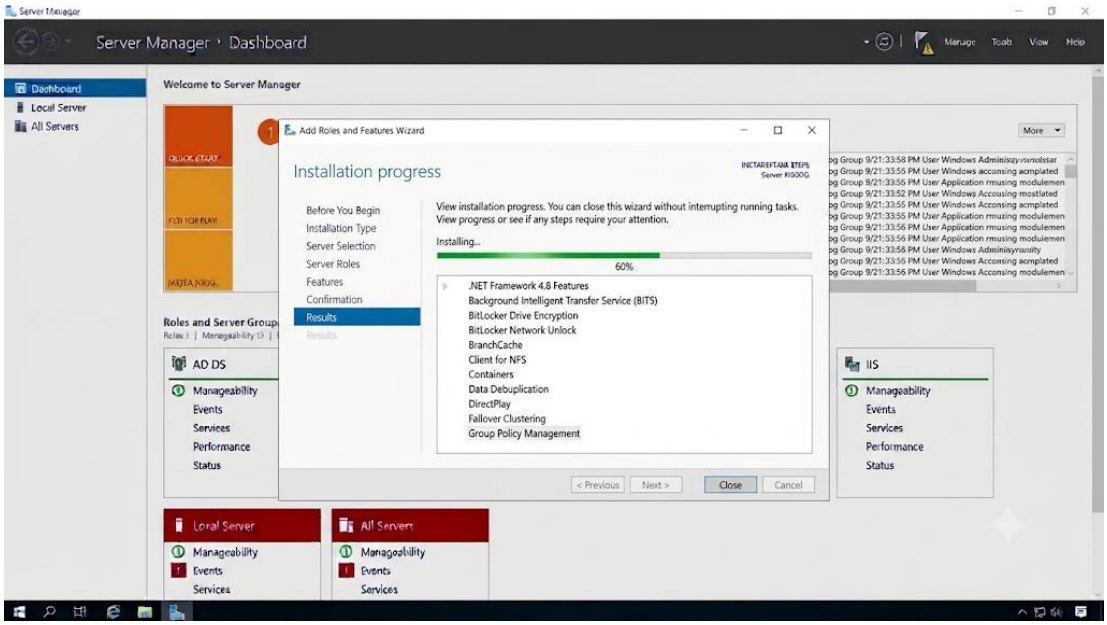
Step 6: Select Features

1. Keep the default features selected.
2. Click Next.



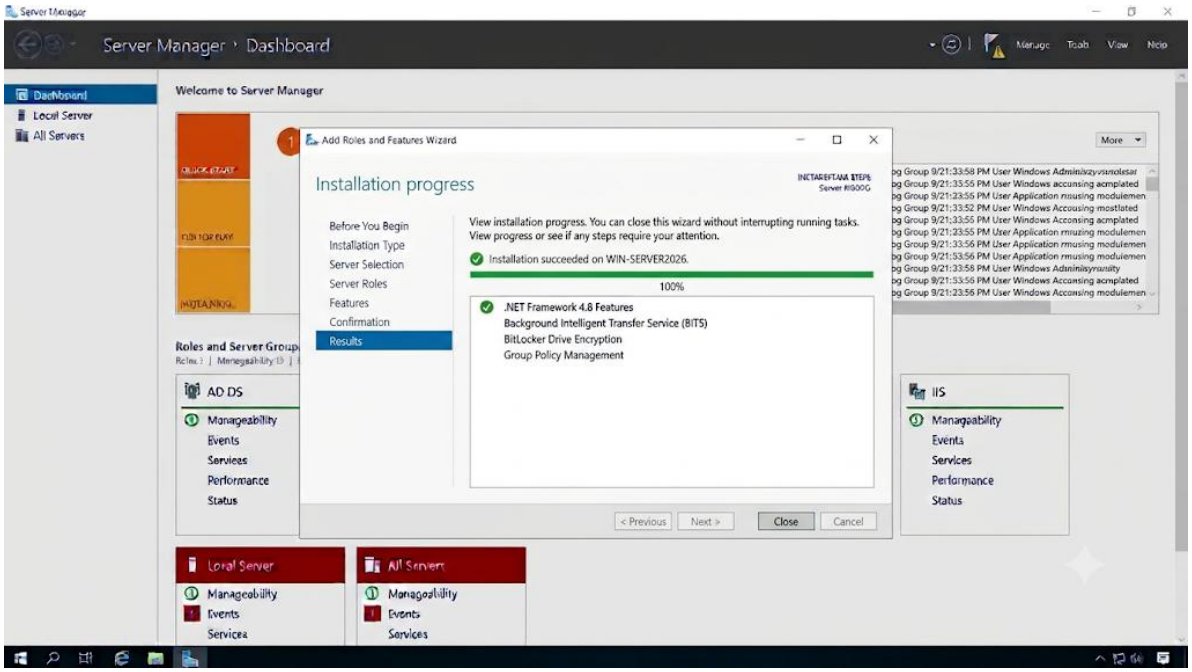
Step 7: AD DS Information

1. Read the information about Active Directory Domain Services.
2. Click Next.



Step 8: Confirm Installation

1. Review the selected roles and features.
2. Click Install.
3. Wait for the installation to complete.
4. Click Close.



3. Promote your Windows Server to a Domain Controller by creating a new forest with a root domain name of your choice (e.g., mycollege.local). List the options you selected during the configuration.

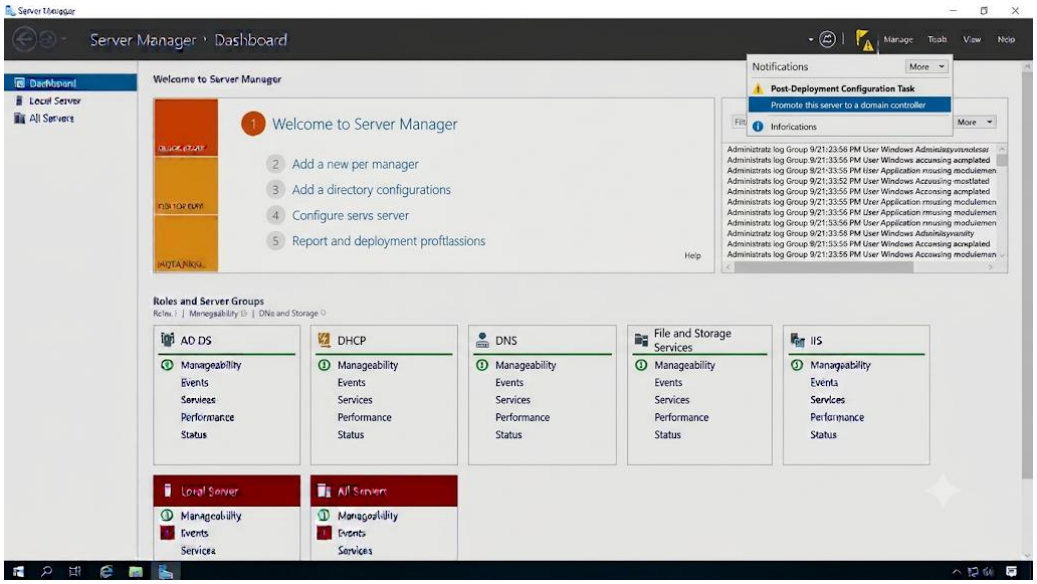
ANS:

Step 1: Open the Promotion Wizard

1. Open Server Manager.
2. Click the notification flag in the top-right corner.
3. Select Promote this server to a domain controller.

Step 2: Deployment Configuration

1. Select Add a new forest.
2. Enter the Root domain name: mycollege.local
3. Click Next.



Step 3: Domain Controller Options

Select the following options:

- Forest functional level: Windows Server 2016
- Domain functional level: Windows Server 2016
- Domain Name System (DNS) Server: Checked
- Global Catalog (GC): Checked
- Read Only Domain Controller (RODC): Unchecked

  Enter a Directory Services Restore Mode (DSRM) password.

Step 4: DNS Options

1. A DNS delegation warning may appear.
2. If this is a new forest without an existing DNS infrastructure, the warning can normally be ignored.
3. Click Next.]

Step 5: Additional Options

1. Verify the NetBIOS domain name.
2. The wizard should automatically generate: MYCOLLEGE
3. Click Next.

Step 6: Paths

1. Review the default database folder.
2. Review the log files folder.
3. Review the SYSVOL folder.
4. Keep the default paths.
5. Click Next.

Step 7: Review Options

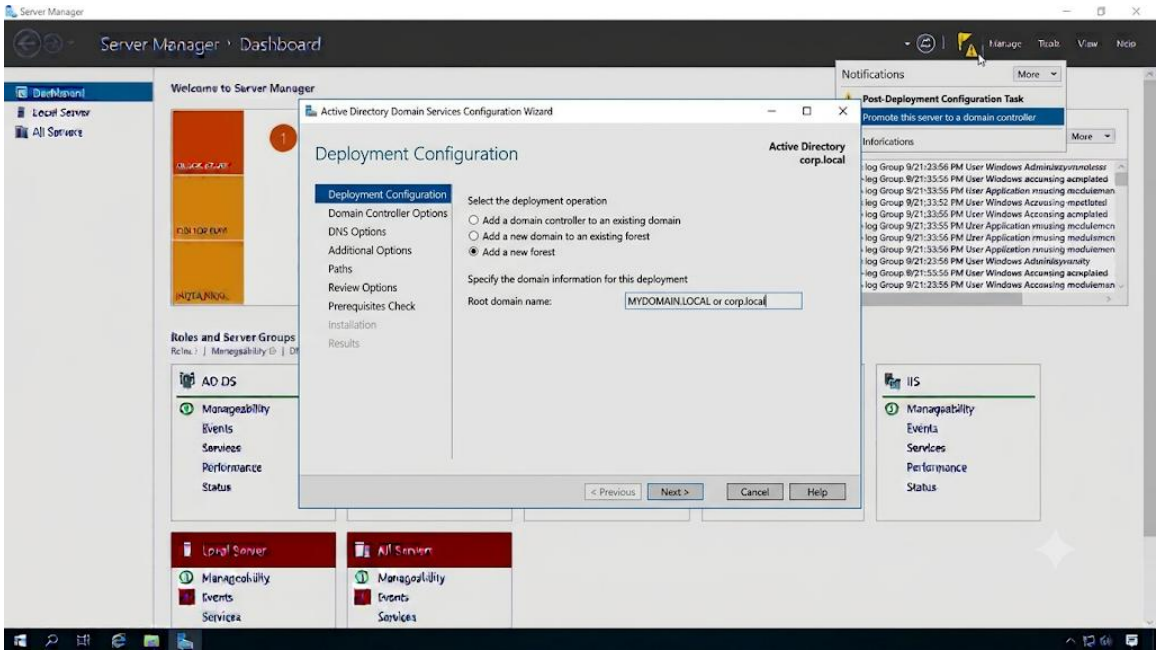
1. Review all the selected configurations.
2. Confirm the domain name and server settings.
3. Click Next.

Step 8: Prerequisites Check

1. Wait for the prerequisite check to complete.
2. Resolve any critical errors if displayed.
3. If only the expected DNS delegation warning appears, continue.
4. Click Install.

Step 9: Restart the Server

1. Wait for the promotion process to complete.
2. Windows Server will restart automatically.
3. After restarting, log in using the domain Administrator account.
   Username: MYCOLLEGE\Administrator
4. Enter the Administrator password.



Domain Controller Configuration Details

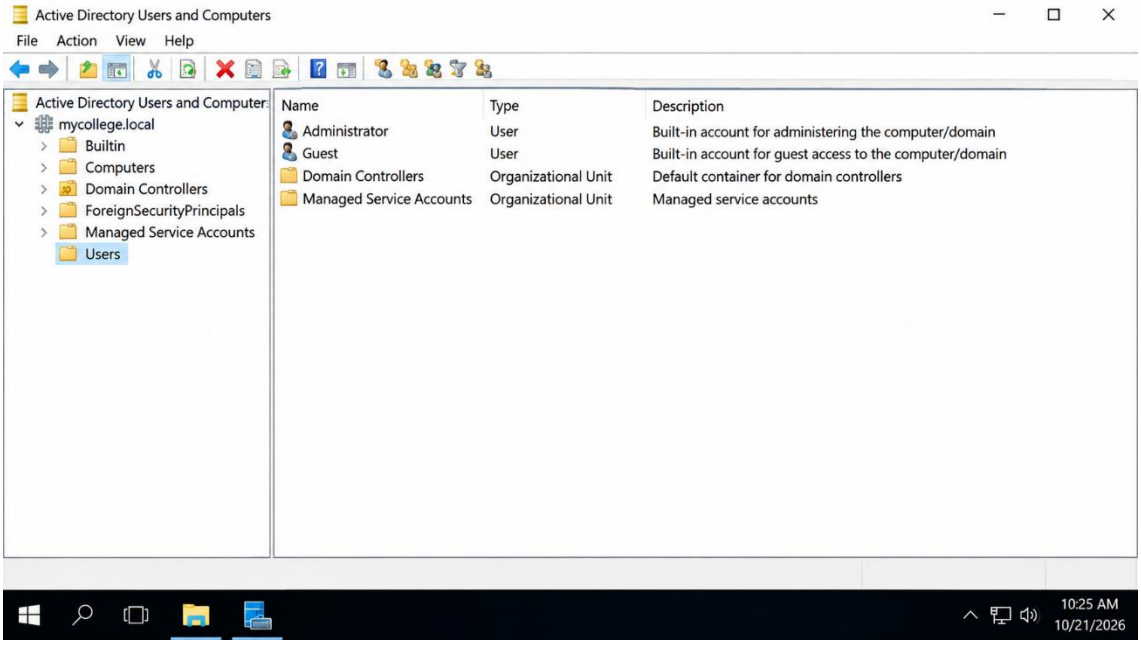
Configuration	Selected Value
Server Operating System	Windows Server 2019
Deployment Operation	Add a new forest
Root Domain Name	mycollege.local
NetBIOS Name	MYCOLLEGE
Forest Functional Level	Windows Server 2016
Domain Functional Level	Windows Server 2016
DNS Server	Enabled
Global Catalog	Enabled
RODC	Disabled
Database Folder	Default
Log Folder	Default
SYSVOL Folder	Default

4. Create two new user accounts and one group in Active Directory Users and Computers. Assign both users to the group and describe how you did it. Hint: Use the 'Users' container and right-click options in ADUC.

ANS:

Step 1: Open Active Directory Users and Computers

1. Open Server Manager.
2. Click Tools.
3. Select Active Directory Users and Computers.
4. Expand the domain mycollege.local.
5. Select the Users container.



Step 2: Create the First User Account

1. Right-click the Users container.
2. Select New.
3. Click User.
4. Enter the details:
5. Click Next.
6. Enter a password.
7. Confirm the password.
8. Uncheck User must change password at next logon (for this lab).
9. Select Password never expires (only for this lab).

Step 3: Create the Second User Account

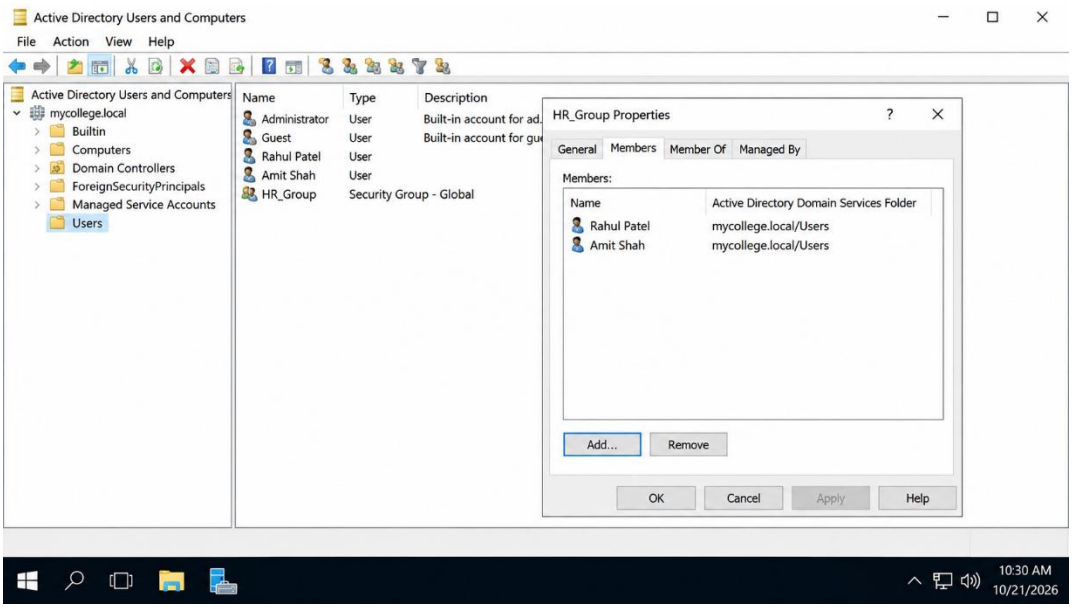
1. Right-click the Users container.
2. Select New → User.
3. Enter the following details:
    - First Name: Amit
    - Last Name: Shah
    - Full Name: Amit Shah
    - User Logon Name: amit
4. Click Next.
5. Enter and confirm a password.
6. Uncheck User must change password at next logon (for this lab).
7. Select Password never expires (only for this lab).
8. Click Next and Finish.

Step 4: Create a New Group

1. Right-click the Users container.
2. Select New.
3. Click Group.
4. Enter the following details:
    - Group Name: HR_Group
    - Group Scope: Global
    - Group Type: Security
5. Click OK.

Step 5: Add Both Users to the Group

1. Double-click the HR_Group group.
2. Open the Members tab.
3. Click Add.
4. Enter the first username: rahul.
5. Click Check Names.
6. Click OK.
7. Click Add again.
8. Enter the second username: amit.
9. Click Check Names.



Active Directory User and Group Details

Object Type	Name	Description
User	Rahul Patel	First user account
User	Amit Shah	Second user account
Group	HR_Group	Global Security Group